

Ajani Martin
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EDUCATION

University of Maryland
MS Mechanical Engineering

College Park, MD
Class of 2028

Lafayette College
BS Mechanical Engineering

Easton, PA
Class of 2026

HONORS

POSSE Foundation Full Tuition Scholarship

December 2022 – May 2026

- A small, diverse group of 10 talented students carefully selected and trained, to serve as a catalyst for individual and community development

Excel Scholar

August 2023 – May 2026

- Research scholarship awarded to students allowing for intellectual opportunities that fuel students' curiosity and passion for topics that interest them

RELEVANT COURSE WORK

Aerodynamics and Compressible Flow, Engineering Gas Turbines, Engineering Thermal/Fluids, Engineering Thermodynamics, Fluid Mechanics, Additive Manufacturing, Mechanical Fundamentals of Mechanical Systems, Dynamics of Physical Systems, Manufacturing and Design, Nature of Engineering Materials

TECHNICAL PROJECTS

EXCEL Research Scholar (Lafayette College Hypersonic Research Laboratory)

August 2023 – May 2026

- Student-driven research on combustion and flame stabilization in supersonic airflow using an aft wall fueled cavity flame holder
- Designed and manufactured a cavity flame holder. A component that sits inside of a Scramjet (Supersonic Combustion Ramjet) whose role is to stabilize the combustion process within the hypersonic/supersonic air flow within the engine
- Conducted research to test the most efficient and suitable location within the cavity to introduce a combustible gas (Ethylene) to the supersonic air flow, allowing for continuous controlled combustion at hypersonic/supersonic speeds
- Conducted tests in the Lafayette College Expansion Tube, a device that allows for controlled testing of vehicles that travel above the speed of sound, through the use of high pressure and combustible gasses

Lafayette College Thermal and Fluids Systems Laboratory

January 2025 – May 2026

- Designed and conducted experiments using common heat transfer equipment, applying principles of thermodynamics, fluid mechanics, and heat transfer to analyze thermal systems and evaluate their performance. Collected and analyzed experimental data to investigate heat transfer behavior and compare results with theoretical principles.
- Analyzed experimental results and communicated findings through detailed scientific reports

Independent Study (ANSYS Fluent and Computational Fluid Dynamics Simulation: CFD)

January 2025 – May 2025

- Utilized ANYS Workbench and Fusion 360 CAD to create scale models to then test through CFD simulation
- Utilized ANSYS Fluent to simulate fluid flow and formation of boundary layers around rocket stability systems, as well as the laminar and turbulent flows around various objects through Computational Fluid Dynamics simulation

WORK EXPERIENCE

Lafayette College Student Researcher | Lafayette College | Easton PA

August 2023 – May 2026

Hours Per Week: 30

- Developed hands-on experience with the Lafayette College Hypersonics Laboratory by learning the proper operation and use of laboratory equipment and systems. Gained an understanding of laboratory procedures, safety protocols, and experimental practices necessary to support hypersonics research and testing

Lafayette College Student Grader / Teaching Assistant | Lafayette College | Easton PA

August 2024 – May 2025

Hours Per Week: 15

- Graded student work for the Strength of Materials course, evaluating assignments and assessments to assess students' understanding of core engineering concepts. Provided supplemental instruction to help students strengthen their understanding of course material, including the mechanical properties of various engineering materials and the analysis of stresses, strains, and forces that materials experience under different loading conditions and applications. Worked

with students to break down complex engineering concepts and apply theoretical principles to practical engineering problems.

Lafayette College Williams Center Equipment Engineer | Lafayette College | Easton PA

August 2025 – May 2026

Hours Per Week: 20

- Supported the successful execution of performances at Lafayette College's Williams Center for the Arts by maintaining and servicing technical equipment to ensure reliable operation throughout events. Coordinated the loading and unloading of equipment and materials for a variety of visiting performing acts, while familiarizing performers and production teams with the venue's facilities and available resources. Worked closely with performers, staff, and venue personnel to anticipate operational needs, address technical or logistical issues, and help ensure that performances ran smoothly for both performers and audiences.

Lafayette College Administration Ambassador | Lafayette College | Easton PA

May 2025 – August 2025

Hours Per Week: 40

- Communicated with prospective and current students to support their transition into Lafayette College by addressing individual questions and providing relevant information about academic and campus resources. Led campus tours for prospective students and their families, providing an overview of the college's campus, facilities, academic buildings, and student resources while answering questions about the student experience. Additionally, conducted specialized tours of the Engineering Department for students interested in engineering, showcasing department facilities, laboratories, and student projects while highlighting the hands-on learning opportunities and resources available to engineering students

JOB RELATED TRAINING

- Received specialized training in the Lafayette College Hypersonics Laboratory, developing hands-on experience with complex experimental equipment and laboratory systems. Trained in the proper handling and management of high-pressure and flammable gases, as well as the operation of high-speed and Schlieren imaging systems used to capture and analyze high-speed flow phenomena. Developed an understanding of laboratory safety procedures, equipment operation, and experimental protocols necessary to conduct hypersonics research safely and effectively.

LEADERSHIP EXPERIENCE

ABC (Association of Black Collegians)

Treasurer: Created and managed a budget of \$10,000, that was used to hold organization events

August 2024 – May 2025

Secretary: Organized all organizational events, meetings, and outreach through detailed scheduling

August 2023 – May 2024

NSBE (National Society of Black Engineers)

August 2022 – May 2026

- *Member:* Engaged in professional development, networking, and community-building initiatives through workshops and organizational events. Collaborated with fellow engineering students and professionals to develop career-readiness skills, expand professional networks, and contribute to a supportive engineering community

LIMS (Lafayette Interdisciplinary Music Society)

President

January 2025 – May 2026

- Reestablished a Lafayette College organization to give students an outlet to express their artistic and musical talents unrestricted by musical taste, ethnicity, religious status, or sexuality
- Organized and supported fundraising events designed to increase student engagement with the surrounding community while raising funds for various organizations
- Collaborated with local businesses and community partners to promote events, strengthen connections between Lafayette students and the local community, and increase awareness of participating businesses

LANGUAGE SKILLS

English

Spoken: Advanced | Written: Advanced | Read: Advanced

TECHNICAL SKILLS

Autodesk Fusion 360 CAD, SolidWorks, ANSYS, CFD, MATLAB, Arduino, LATEX, LaVision Davis, Microsoft Excel, Microsoft Word